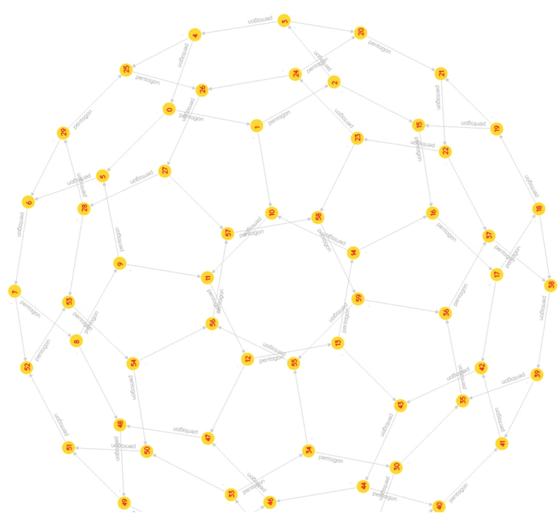
CS 2C Syllabus - Winter 2024

Data Structures and Algorithms using C++

"You cannot beast mode code your way through this class" - John Vicino

Learn Computer Science like in no other place



Purty Pitcher created by Dylan Hendrickson, Spring 2023
Final homework submission for CS2B
https://www.reddit.com/r/cs2b/comments/14b909z/quest_9_purty_pitcher
https://www.reddit.com/r/cs2b/comments/14c01cl/dylans_jelly_soccer_ball

CS2C

My name is &. I'll be your teacher in this class.

I've significantly redone the structure of the class this quarter, largely in response to useful feedback from past students and what some of you have asked for. Here is what is new, in case you were expecting the old kind of syllabus:

- Weekly action plans (given in Canvas)
- Weekly graded reflections posted by you in the discussion forum
- Optional 1-1 weekly exam/interview with me or a proctor.
- 1 quest due per week (not 2 every 2 weeks like last quarter)
- Live-coding opportunities with extra credit projects

About this class

If you're new to this kind of class, you should note that it is different from other traditionally taught classes. I find more students tend to like it this way than not. I think students learn better too. However, it takes a little getting used to. It is like a **bootcamp** for those who want to learn Computer Science principles through C++, one of the most powerful programming languages.

Students will be required to do their homework assignments at the Genius Bootcamp. Those who are enrolled in this class are auto-enrolled in this bootcamp for free. I have arranged to get notifications and access to all of your code submissions, and for your scores to be transferred to Canvas on their due dates. This bootcamp is an incredibly fun and supportive community experience to learn CS in. However, you should not expect answers or hand-holding by anyone here either (as these are discouraged).

In many classes that I've seen, and in ones I have taught in the past, students typically compete with each other. There is a tremendous amount of secrecy around the material each student knows, and requests for help are made and received in private. In the end, I found that many students didn't actually learn what the course promised to teach them. However, they had an easy *time* in the class because solutions were easy to get, easy to get accepted, and the work wasn't demanding.

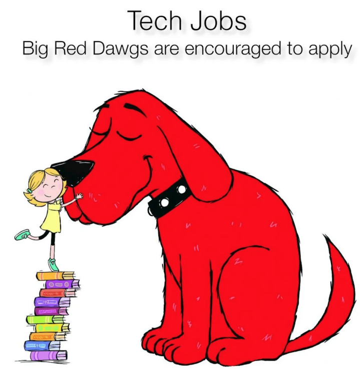
In contrast, in this class, you will likely find yourself struggling. You will be frustrated. You will want to give up. But thanks to incredible support from our unique CS community that YOU get to be part of, you will feel happy that you didn't. Read the reflections and advice for future students from those who already did it.

You should budget enough time for this class in your schedule. If you're enrolled in 15 units, for example, and you're doing one other time-consuming class, say in Math, then you should ideally pick your third class from an area you are not intending to major in (e.g. Ceramics, Acting, etc.). This will give you the time to focus on the STEM subjects aligning with your major, while providing a great and relaxing non-STEM class experience. Foothill counselors and academic advisers are trained in helping you with this.

You should ideally start questing and participating in the bootcamp as early as possible after you get this syllabus in your hands. More details on this later in the syllabus.

Here is how your class homework problems align with Genius.

- CS2A Homework gets you through the BLUE level of the bootcamp (Beginner)
- CS2B Homework gets you through the GREEN level of the bootcamp (Intermediate)
- **Belief in yourself**, courage, patience, and perseverance make you a RED DAWG (Pro)



Start participating in our incredible and lively community right away.

CS 2C Readiness Self-Calibration

This is a **challenging** class according to most former students. How do you know if you will be successful? Try this:

In this class, there are two required technical tasks to test your preparedness AND prepare those who need a refresher.

- 9 BLUE quests = Beginner level Genius (*essentially ALL the homework of CS2A*) is due on day 2 of week 1). **This is midnight, Tue Jan 9**. You only get 48h to complete what CS2A students get 3 whole months for. Why?
- 9 GREEN quests = Intermediate level Genius (*essentially ALL the homework of CS2B*) is due on day 7 (Sun of week 1). **This is midnight, Sun, Jan 14**. You only get 5 days to complete what CS2B students get 3 whole months for. Why?

Even though you may already be proficient in these concepts or have passed CS2A and CS2B (or their equivalents) elsewhere, you still need to complete these quests. I allocated grade points to them commensurate with the amount of effort they should take CS2C beginners.

If it takes you longer than 2 days to complete all the BLUE quests, or longer than 5 days to complete all the GREEN quests, you should seriously reevaluate if you want to do CS2C this quarter or later or review the amount of effort you are willing to put in each day.

Important note: To keep your self-evaluation meaningful, you must **NOT** refer to ANY past-subreddit posts, nor search for solutions online. You may converse freely with any *current* quester in the appropriate subreddit. Each quest is designed to be solved using only the tools you should already be familiar with when you reach them (plus your own creative thinking). *This is the way the very first batch of questing students had it, and I don't believe that you are less capable than any of them.*

What does the data say?

Students who complete GREEN 9 by day 10 have an 73.9% chance of a A- grade or better

One by-product of this necessary exercise (if done sincerely) is that you will get a taste of the pace of work required for this class. You cannot return to the class from time to time to make progress. It has to become a central theme of your life for the remainder of this quarter and you must either be programming or thinking about programming at least 4+ hours each day. To make it more worthwhile, my participation points award system is tuned to reward predictable periodic forum behavior and non-bursty progress.

Administrative stuff

CS 2C continues where CS 2B left off. Students already comfortable with intermediate-level C++ programs will have an opportunity to play with and master important data-manipulation algorithms.

This course provides a systematic treatment of advanced data structures, algorithm analysis and abstract data types in the C++ programming language. Coding topics include building ADTs on top of the STL templates, vectors, lists, trees, maps, hashing functions and graphs. Concept topics include searching, big-O time complexity, analysis of major sorting techniques, top down splaying, AVL tree balancing, shortest path algorithms, minimum spanning trees and maximum flow graphs.

A strong familiarity with algebra and good written English comprehension skills are both advisories. Please arrange tutors/helpers right away if you need assistance with either. Contact the STEM Center or the TLC.

Course outline and SLOs

You can access [the official course outline of record for all CS courses here]. Student Learning Outcomes for this course are:

1. A successful student will be able to write and debug C++ programs involving advanced data structures such as Lists, Trees, Graphs. They will be familiar with the use and implementation of algorithms for balancing binary trees, creating splay trees, minimum spanning graphs, finding the shortest path through a graph, and maximum flow through a network. They would also be familiar with the most common sorting algorithms and know the advantages and tradeoffs of each.
2. A successful student will be able to reason about the running time and derive properties of computer programs using precise mathematical terminology. Specifically they will be conversant with the Big-O notation and be able to craft efficient algorithms using the appropriate data structures to solve non-trivial computational problems.

Canvas and required tasks

Students at Foothill College doing this class for credit should use Canvas to coordinate some activities and take online exams. Most of the rest of our work will happen at other public locations, including youtube, zoom, quests, reddit, etc.

Start your adventure in Foothill's class by completing the following tasks (the first two are **required**). The Canvas site for this quarter should be open if you're reading this.

1. posting an introductory note about yourself in Canvas (You can simply reply to my own introductory post if you prefer)
2. introducing yourself in the subreddit under the Genius Flair
3. scoring at least 90% on the syllabus quiz (in Canvas). This doesn't need knowledge of CS or c++.

☆ These **must be completed before the end of the first Friday (Wednesday for summer) of the quarter** to prevent being dropped.

Course Workbook

Your homework problems are designed to cover a well-spaced selection of topics in the syllabus (and in your Canvas Action Plans). These are the LAST 9 quests of the official Genius workbook - The Enquestopedia.[1]

[1] The Enquestopedia is just a high-level reference of what is to come. You will get Foothill-tailored versions of the quest specifications (slightly different from the bootcamp workbook) as and when you complete each quest.

Assessment

If you're doing this course for kicks, or other fun reasons, you can skip this section.

If this course is offered for a grade, and you are taking it for a grade, then, your final grade will be based on fun programming quests[2] (homework), exams, and conceptual understanding/ability to help other programmers (forum).

Worksheet

In the worksheet below, activities marked with an asterisk (*) are **required** activities. This means that if that activity is not completed then nothing below it will count towards your grade. *Pupping* a quest means progressing until you get the password to move on. *Dawg* stands for *detailed attentive work given*, which is proxied by collecting the stated number of trophies for that color).

How to calculate your grade

What	Scaled from	Scaled to
Syllabus Quiz*	20	1%
BLUE PUP*	10	5%
BLUE DAWG (>= 191 Blue trophies)	4	4%
GREEN PUP*	10	10%
GREEN DAWG (>= 241 Green trophies)	5	5%
RED PUP	265 (trophies)	25%
RED DAWG (>= 265 Red trophies)	5	5%
Midterm + Final (20+40)	60	25%
TEN Weekly Reflections (Weeks 1-11)	100	10%
Class participation and final reflection	10	10%
TOTAL		**100%**

Then use the absolute grade map below:

For an	A+	A	A-	B+	B	B-	C+	C	D	F
You need (%)	97	91	88	86	80	78	75	67	60	< 60

[2] More on quests in Appendix A.

How to do your weekly reflections

Each reflection can net you anywhere from 0-10 points depending on its quality. You will need to make at least 10 reflection posts in the subreddit under the *Reflections* flair over the course of the first 11 weeks of this class. They are due on the first 11 Sundays. Each is worth up to 10 points, and together contribute 10% to your final grade. Thus, no reflections implies a maximum score of 90% (or borderline B+/A-) no matter how well you do in the other areas.

This NYT link, shared with me during the last quarter, gives lots of tips on how to prepare good reflections. You can use it as a template or a workbook to create a new one each week.

Your weekly reflection should also showcase your forum contributions during that week. Focus on at least some of the following:

1. What do you know (in CS/C++) this week that you **didn't** know well last week?
2. Are you stuck with your homework? Frustrated? Too shy to ask for help?
3. Did you make an exhilarating breakthrough this week on something that had you stuck for ages?
4. Did you share any useful or interesting information on our forum?
5. Did you help anyone?
6. Are there classmates you'd like to thank for helping you?

NOTE: You may be collaborating on extra-credit or other projects with your classmates. But your project updates should not take the focus of your reflection. Maybe a line or two at most.

At the end of the quarter, you can summarize your 10 weekly reflections along with quarterly observations into your final report/reflection (worth 10 points on its own).

Weekly reflections don't need to be long. You will find that a couple of paras suffice for most weeks.

You cannot make up points for missed reflections except by passing the 1-1 interview within a day or two of missing it.

Final Reflection (Week 12)

The final reflection should be a post in the subreddit (E.g. Final Quarterly Reflection by XYZ) summarizing your experience during the entire quarter.

To score high on this, your report should be prepared as if it is a portfolio for a potential hiring manager who may look you up and review your reddit posts to gauge your level of expertise and ability/willingness to learn new things. You can use this as an opportunity to build a compelling portfolio of technical musings and discussions you might want to present to someone. This is when you'll be glad you wrote weekly reflections because you can simply use them as the raw material from which you can create this summary final reflection.

It should contain links to *selected* posts and comments (with short descriptions) spanning the entire quarter, not just all made in a few weeks. I also recommend you use your post to leave comments and advice for future students.

A good final reflection has a selection of links and comments for the following types of content. See examples on our sub.

- Posts that sparked or contributed to rich discussion (lots of good comments)

- Posts that say something insightful
- Posts that offer helpful and kind advice to other students who may be struggling
- Posts over the entire duration of the quarter, rather than in spotty bursts once in a while
- Posts that leave sound advice and encouragement for future students

Avoid posting a list or table of links, or listing trivial posts, comments and answers. Ideally, all your selected links had something valuable to ask or say and you have something to say about them now. Use the subreddit flair Reflections for this post.

Also use this post as an opportunity to thank all your classmates who stood by you during the quarter and pulled you out when you were stuck. It should essentially be a good summary of your 10 weekly reflections. Quests need to be pupped in sequence before their freeze dates. However, once pupped, they can be dawged any time before Friday of W11. Email me when you're sure you have dawged a RED quest.

If you absolutely abhor having to post a weekly or final reflection in our subreddit, you have an option (as does anyone who gets a 0 for a particular week's reflection). You can schedule a 1-1 examination with me, during which you will solve some programming problems live while I'm able to observe it - you won't be able to ask me for clarifications during this test. Recorded 1-1 exams will be shared on Canvas for everyone's benefit.

You can go to the respective subreddit (r/cs2a, r/cs2b or r/cs2c) and simply search for posts with the word "participation" or "report" in the title. You are bound to find many fine examples.

Exams

Exams are objective style and will be administered via Canvas. You will typically have a window of time (18+ hours) during which you can begin these exams. But once you begin, the current version of Canvas does not allow you to *pause* your exam and come back to it. The 1h (or 2h for final) timer cannot be stopped once you start it, until you hit finish.

They can be taken anywhere you get a decent Internet connection. But I don't recommend taking them on mobile devices. I also suggest that you don't actually try out code-fragments from your question sheet in an IDE.

Extra Credit Work and Internship

Extra credit work may be offered to selected students from time to time. Normally, you need to have no overdue quests to be offered an extra-credit project. I may offer SW Engineering Internships at Nonlinearmedia to a select few RED DAWGS with an awesome reddit profile.

Virtual Catch-up Circles

These zoom meetings are TREMENDOUS collaborative opportunities and have provided endless hours of support, both technical and emotional. However if you attend these meetings only to get unblocked when you're stuck, you'll probably not score high on participation. Use the meetings to generate interesting ideas and insights which you can then post about in the forum (these kinds of posts get most points). Once the class is up and running, we'll schedule Virtual Catchup meeting times on Zoom.

Any class-related discussion that may be generally useful will be made available as transcripts or recordings via our Canvas class site for all students. This includes 1-1 meetings. Make sure you are ok with this policy before enrolling.

Communication

Please use our sub for any question or comment that relates to the quests and CS concepts (except questions of a private nature). If you have a confidential question (grades or registration) you can email me.

Try to meet with each other after class (even if virtually), set up private study and programming groups and work on independent (non assignment) programming challenges outside of class. I'll give you a few interesting challenges from time to time. Some of these may earn you extra credit.

You can reach me via messaging in Canvas, Reddit or by email. While on campus, my room number is `0x113d` (in hex). In week 1 of CS2A, you will learn how to decode that into decimal.

Don't use Discord. Things you share on discord are not visible to us and cannot be used as proof of anything required in this class.

Infinite Office Hours!!!

Well, within reason, ofc.

One time I happened to be in my office at 2AM. Someone knocked on my door.

> "Who is it?" I asked.
>
> "Quick question prof" said a female voice.
>
> I opened the door. "Hey sorry. My office hours are at 10am. See the sign?" I pointed at the print out I had stuck to my door. It clearly said 10AM - 11AM in **BIG BOLD BLACK** letters.
>
> "Yeah" she said. "I read it. But I'm a binary janitor. So I waited up until now to come here."
>
> "That's cool. You definitely deserve an answer" I said. "What's your question?"
>
> "Would you like me to come back later to empty your trash can?"

This quarter, I've decided to do away with this confusion. I have open office hours ANYTIME (virtual). I am very flexible in being able to adjust to your availability. But you have to pre-arrange it with me via email first.

I will likely be able to point you in the right direction for further experimentation. But I will not look at your questing code directly, nor debug it for you. Nor can you ask anyone else to do it (honor code).

Schedule and Homework Deadlines

Programming, like all art, is not a 9-5 job. Sometimes you're on a roll and killing it. Other times, not so much.

I know how it is.

So you can quest at your own pace. However, note that this quarter there is a ***late fee of 5 trophies*** for each quest that is submitted past its freeze date. These points will be subtracted from the final trophy count for the corresponding quest, but will not be less than 0.

Here is a suggested study/work plan, which aligns closely with the Weekly Plans you will find in your Canvas Modules.

Week	Read References	Complete	Notes
1	Algorithms and ADTs review	All BLUE ALL GREEN	
	Algorithm analysis	Mystery Quest 1	Q1 Freezes Sunday night
3	Time complexity and Big-O	Mystery Quest 2	Q2 Freezes Sunday night
	General trees (and BSTs)	Mystery Quest 3	Q3 Freezes Sunday night
5	AVL Balancing and Splaying	Mystery Quest 4	Q4 Freezes Sunday night
	Review/Midterm (Canvas)	Mystery Quest 5	Q5 Freezes Sunday night
7	Hash tables Quadratic probing	Mystery Quest 6	Q6 Freezes Sunday night
	Sorting	Mystery Quest 7	Q7 Freezes Sunday night
9	Priority Queues, Heaps, Heapsort	Mystery Quest 8	Q8 Freezes Sunday night
	Dijkstra's and Kruskal's algorithms		Aim for 50% of Q9
11	A Maxflow algorithm	Mystery Quest 9	Q9 Freezes Sunday night
	Final Exam (Canvas)		

You only get 1 week to complete each quest - hard-deadlines in CS2C

Every week, give yourself ONE or more topics to study and one or more programming quests to complete (2 in summer). Anything not clear should be clarified via discussion in the forum, ideally during the appropriate week (See table above).

Make sure you have adequate time to code, experiment, revise and recode EVERY week. This takes a LOT longer than many students estimate. Also see: [The bell rings at 11:59](#).

Finally, note that you cannot leave a hole in your questing trail. Your trophies for Quest-n only count after you have at least pupped all quests < n-1

Extensions

Extensions don't make sense because the quests are self-paced. You just have to complete each by their "freeze" date to get credit. After their freeze dates, you should still complete them, but not for credit. There's a LOT of time to complete these quests even if you have to take some breaks. So, there is no need to ask for extensions.

Disability Resource Center

Foothill College is committed to providing equitable access to learning opportunities for all students. Disability Resource Center (DRC) is the campus office that collaborates with students who have disabilities to provide and/or arrange reasonable accommodations.

If you have, or think you have, a disability in any area such as mental health, attention, learning, chronic health, sensory, or physical, please contact DRC to arrange a confidential discussion regarding equitable access and reasonable accommodations.

If you are registered with DRC and have a disability accommodation letter of accommodations set by a DRC counselor for this quarter, please use Clockwork to send your accommodation letter to your instructor and contact your instructor early in the quarter to review how the accommodations will be applied in the course.

Students who need accommodated test proctoring must contact the DRC immediately if they cannot find or utilize your MyPortal Clockwork Portal. DRC strives to provide accommodations in a reasonable and timely manner. Some accommodations may take additional time to arrange. We encourage you to work with DRC and your faculty as early in the quarter as possible so that we may ensure that your learning experience is accessible and successful.

To obtain disability-related accommodations, students must contact the Disability Resource Center (DRC) as early as possible in the quarter. To contact DRC, you may:

Visit DRC in Building 5400, Student Resource Center (physical visits suspended during college closure).

- On the web: http://www.foothill.edu/drc/
- Email DRC at drc@foothill.edu
- Call DRC at 650-949-7017 to make an appointment

Important Dates

For a list of important dates for the winter quarter, see the official college page here.

Appendix A - Questing

The Genius Bootcamp

The Genius Bootcamp is a supportive and fun community of CS and C++ learners from all over the world. You are auto-enrolled in this bootcamp and you can proceed to complete it for free after you finish CS2B. In fact, you can even get credit for the BLUE and GREEN quests if you decide to enroll officially in my CS2C next quarter.

At this bootcamp, you are bound to have company from Genius questers who are struggling with the same kinds of things as you. You may find that your posts in the subreddit elicit help or responses from bootcampers not in your class.

Please introduce yourself in the Genius Flair of the red subreddit. After that, your interactions will mainly take place in the blue subreddit, r/cs2b.

Mystery Quests (capped at **265** for CS2C this quarter)

You will solve these at the public questing site (https://quests.nonlinearmedia.org). If you're new to questing and haven't completed the BLUE level yet, type in the password "*A Tiger named Fangs*". You should hear a whooshing sound. Hit return to enter the first BLUE quest. The spec of each quest (detailed description of what you need to do) will be shown when you click on the name of the quest at the top. The first 5 quests are almost trivial, and are meant to get you warmed up. If you're in 2B or 2C, you should aim to finish all 5 in a SINGLE sitting at your computer.

These quests are meant to be solved **in sequence.** Each quest will give you a certain number of trophies. You can check your total trophy count at any time by visiting your personal scoreboard at the /q site (It will be wiped on the 1st of Jan, Apr, Jul, and Oct). *It will show you all your trophies, but only the BLUE and GREEN ones count for this course.* Your secret handle is your Student ID

The quests are set up such that the password to each quest is given out upon scoring a certain number of trophies in the preceding quest. However, I found that a few students were getting stuck in the lower numbered quests pounding away at them to eke out every remaining trophy before moving on, even though they had already earned the password. This is a bad strategy. Resist your temptation to hack more, and keep moving when you get a password. P*upping*[3] a quest is sufficient to move to the next quest. You can always come back to polish your previous quests when you have free time before the final week. You get about one week on average before each quest freezes (only 3 days in Summer). After this, the trophy count for a quest will be docked by 5 trophies (minimum = 0).

> **What does the data say?**
>
> **Students who don't complete Quest 4 by week 6 have an 61.7% chance of a C-grade or worse**

At the end of the quarter, your RED trophy count will be capped at **265** and fed into the grade crunching system. If you spend a lot of effort getting up to high numbers by the time you get to Quest 7 already, then you'll be close to getting burned out right in time for two of the funnest quests of all. So plan your time and effort wisely. It's not like your old quests are going to disappear when you move on.

[3] Progress Until Password

In order for rewards from a quest to count towards your total, you must have completed all previous quests. If you leave a hole in your trail of completed quests, then your total reward earnings is the sum of all rewards you earned before the first incomplete quest.

Pupping and Dawging quests

Solving minis until you get the password to move on from a quest is called pupping it. This is what I recommend you do in your first pass through the questing trail.

Dawging it means you review the code and the spec in detail and try to get all available trophies. I don't think anyone knows how many trophies there are per quest, but it's usually easy to tell if you've got everything obvious. If you Dawg all the quests you should get at least 265 trophies (the cap).

What is a Quest Freeze?

A freeze is when I will review your submissions and transfer scores from a particular quest into Canvas. If a quest has frozen, you still have to complete it in order to move forward into the next quest, but you will have to pay a late fee of 5 trophies for this quest.

Bummer… Yes? If I were you I would try my best to avoid that situation by staying on top of things and taking this class ***seriously***.

Dawg points are **All OR NOTHING**. That means, "264 of 265 RED trophies" is not worth any points until one more trophy is won. No exceptions.

Bugs in your code?

Getting your code debugged by someone else is NOT allowed. That includes me, tutors, teachers, friends, enemies and relatives. Debugging your own code is an essential skill that aspiring programmers must learn and enjoy - Yes, enjoy!

Of course, I can't police this. But your enrollment in this class signifies acceptance of this condition (in addition to being bound by Foothill's Academic Integrity Policy). You cannot send your code to me, a tutor, or someone else and ask them what the issue is. What you can do is:

1. Explain (in our **subreddit**) what you're trying to do
2. Describe in English the steps you think might work, or if you have no idea and would like someone to explain the requirement better.
3. Or describe the behavior of your program and ask why it is not working as expected.

Sometimes it is also ok to post some code on our **subreddit**. Mostly, exercise good judgment regarding what can be shared. You want a fun and fulfilling learning experience. The best way to get it is to keep it fun and fulfilling for everyone. You wouldn't give away a movie's ending to a friend who's going to watch it. Why give them the solution to a problem when they can feel good finding it themselves?

Appendix B - Reddit Recommendations

Subreddit username

Your reddit avatar name **must** start with your first name (as on Canvas) and an underscore, followed by your initial (or full last name) + some optional digits (example: *ramanujan_s1729*). If you already have a conformant reddit username from your CS2A or CS2B enrollment, I highly recommend you use it so that people who click on the username can see your contributions over the entire series. Only posts and comments made by usernames matching the above format are eligible to be linked into your final report.

Don't say anything in the forums that you'll end up regretting later in your life. OTOH do try to let your natural genuine curiosity shine through for others to seek out in a sea of wannabe programmers. Maintain your profile on our subs as you would if you were a professional and it will free up a lot of your time.

I will try to remove posts that I deem (in my subjective opinion) to be a liability to your future self. But you can't rely on it. Best to be helpful, courteous, informative and only post useful and interesting observations without overtly giving the answers away. They usually have more lasting value.

ANY user anywhere in the world can quest and post/discuss in our subreddits. So you may see posts and replies by users with anonymous names like *coding_lion*, *bat_girl* and such. All posts are subject to the same rules like *Johnny be good*, but only the ones with avatar names matching the spec in this syllabus will get participation credit:

Can you review past subreddit posts?

No. Access to past posts in the subreddits **AND to posts giving tips/hints from current students** is off-limits to current students (also recommended for current non-student questers).

You are expected to use the subreddit to communicate with your current questers, and to demonstrate both your conceptual understanding and willingness to help others technically.

After you successfully complete a quest on your own (with possible contemporary help), you can refer to all past posts for that quest before you post your insight post.

IMPORTANT: Even though our subreddits contain content created by past students and other bootcampers, you must NOT review any past content on a quest - honor code. You can view and comment as much as you want on any content for a particular flair after you have *dawg*ed that quest. Don't publish tip sheets or cheat sheets. Do not share actual quest code.

Appendix C - Resources

Text recommendation

My first resource suggestion is the book: *Data Structures and Algorithm Analysis in C++*, any edition ≥ 2nd, by Mark Allen Weiss.

Then there is, of course, the Internet. You will need to search various sites like Stack Overflow, reddit, YouTube, etc, for the keywords of the week. You can use your weekly reflection to share links you found useful (along with a short summary of what you found useful in them).

If in doubt about anything, you can ask your question on the subreddit even while you do your own research, and if no one responds to your question, you can close it out yourself by answering your own question after a few days! Very valuable for reflection posts, I think.

The way to study in this class is:

1. Identify the key terms you want to search for (e.g. "*templates*", "*spanning tree*", "*big-o*", etc.)
2. Look in the index of your text to locate the corresponding topics in the book
3. Search the internet to locate helpful other resources
4. Share your thoughts and resources you find on our forum and participate in community discussions around them.

I may offer guest lectures on some of the topics students have found challenging in the past. I will announce on Canvas when these happen and you are expected to attend them (or excuse yourself by emailing me).

Finally,

> *As always, hit our **sub**, when in doubt.*
> *Actually, that's not quite it.*
> *When in doubt, try it out.*
> *If you still don't get it, then hit our **subreddit** (to ask - not to look up)*

Bottom line

Computer science is a **hard science**, not a soft science. A significant investment of your time and effort will be required in this class. To succeed in CS2C, you must expect to code at least 2 hours EACH and EVERY single day for the next 12 weeks. Make sure your schedule allows it before you start. Now smile, and get ready to get wasted (in a good way). If you apply yourself sincerely, you will learn a REALLY USEFUL skill for a happy life.

Happy Hacking!

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